Summary on "A Minimum Interference Routing Algorithm"

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Abstract: Minimum Interference Routing is instrumental to MPLS Traffic Engineering under realistic assumptions of un-known traffic demand. This work presents a new algorithm for minimum interference routing, called Light Minimum Interference Routing (LMIR). This algorithm introduces a new approach for critical link identification that reduces the computational complexity. Results, derived via simulation, show that LMIR is precise and has indeed a low computational complexity.

Introduction:

Path selection for a given source-destination (SD) pair traffic may follow different criteria. Minimum interference routing algorithms have attempted to reduce blocking probabilities by finding edges which minimizes the maximum flow reduction between other SD pairs.

This is a paper which introduces us with an algorithm named "Light Minimum Interference Routing". This algorithm tries to optimize the resources and algorithm tries to use in a lower complexity than other routing algorithms.

Light Minimum Interference Algorithm:

Existing all algorithms about minimum interference has max-flow type solution. The fastest max-flow algorithm is Goldberg's algorithm. This Goldberg's Algorithm has a time complexity of **$O(\min(V^{2/3}, E^{1/2}).E.\log(V^2/E).\log(U)$** with |V| vertices, |E| edges with capacities in the interval of [1, U]. In this paper there is a better algorithm which can be used more efficiently than Goldberg's algorithm.

LMIR attempts to find the K paths with the lowest capacities among all SD pairs in order to determine the critical edges. After finding K paths, a calculated new weights [called w(l)] are assigned to all of their edges. Then Dijkstra's algorithm using w(l) as weights is executed for selecting non-critical edges or edges with low criticality. Here the Dijkstra's algorithm is little modified and here it is named '**LowestCapacities'**.

LMIR algorithm includes 'LowestCapacities' which has a complexity less than $O(V^2)$, Here they considered $O(V^2)$. Each pair executes algorithm K times so complexity is $O(K.\ V^2) = O(V^2)$. Now they showed that for both dense and sparse graph network, the critical edges detection in LMIR is less expensive than in the algorithms that use maxflow. They showed LowestCapacites = O(Goldberg's maxflow).

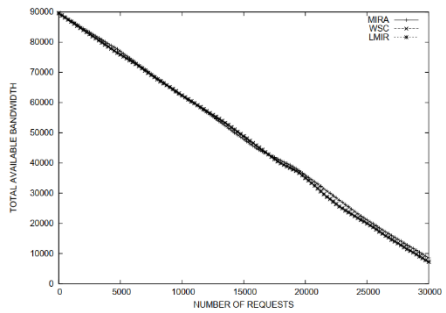
The proper choice of K is very important here. As it is not possible to establish an optimum value, they established an upper bound for K, here upper bound is defined as $K(\theta)$. To compute the $K(\theta)$, **only one** maxflow algorithm is needed to be executed, but in the other algorithms, a max flow algorithm had to be computed **in every request** from node u to v.

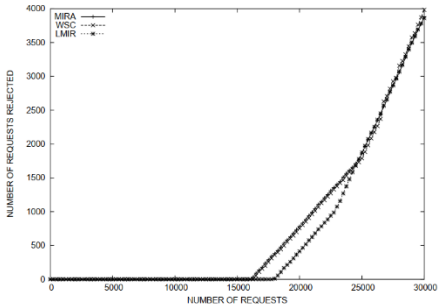
Numerical Examples:

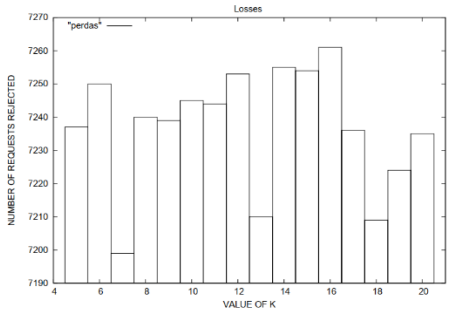
They showed LMIR produced flow reduction rate vs other algorithms' flow reduction rate with respect to number of requests. Other algorithms are Minimum Interference Routing Algorithm (MIRA), Minimum

Hop Algorithm (MHP), Widest Shortest Path Algorithm (WSP). They also showed some numerical examples of the number of requests rejected with respect to number of requests. An example of 50 nodes is taken of both dense and sparse networks, and all the edges with capacity of 1200 bandwidth units.

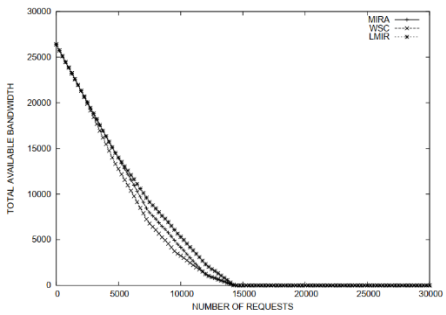
For **dense** network, the behavior of WSC, MIRA, LMIR is quite similar up to 20000 requests, after that MIRA achieves slightly smaller reduction in the maxflow compared to the others. And if we observe the number of rejections, the first rejection of LMIR occurs after 17500 requests, but the other algorithms' occurring rejections starts around 1600 and rejects a larger number of requests than LMIR in a range of [16000, 25000]. So LMIR performed better. Graphs:

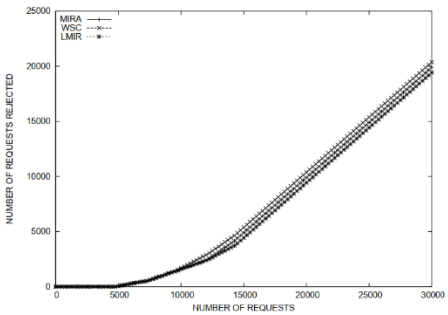


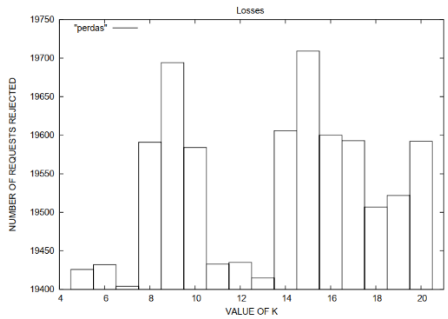




For **sparse** network, we see that, Reduction of maxflow is smaller in LMIR than other algorithms. After 4000 requests, WSC presents larger reduction than LMIR and MIRA. Overall, the LMIR works better here. And if we observe the number of rejections, LMIR rejects a smaller number of requests (19,500), where others reject much more such as MIRA (20,000), WSC (20,500). Graphs:







The value of K is to be chosen perfectly, LMIR quality depends on the choice of K.

Results:

In dense large network, LMIR works similar to other like MIRA and WSP. But in sparse network, LMIR works slightly better than the others.